

Implementation of a Raytracer

Master - Lab Course / Praktikum (6 ECTS)

- Modelview transform
 - Homogeneous notation
 - Placement of objects and camera
- Ray-object intersections
 - Implicit surfaces, e. g. quadrics (e. g. sphere)
 - Parametric surfaces, at least triangles
 - Axis-aligned boxes AABBs for acceleration purposes
 - Optional: CSG operations, i. e. combinations of objects with Boolean operators
- Shading
 - Phong
- Shadows
 - Point light
 - Directional light
- Sampling
 - Random
 - Stratified
 - Poisson or Best-Candidate
 - Optional: Hammersley or Halton
- Reconstruction
 - Box
 - Mitchell
- Intersection Acceleration
 - BVH or uniform grid or kd tree
- Test scenes
 - Implicit surfaces
 - Triangulated surfaces

Master - Team project / Master project (16 ECTS)

- BRDF
- Monte-Carlo integration
- Texture
- Area light
- Ambient occlusion
- Transparency
- Depth-of-field

Literature

Kevin Suffern, Ray Tracing from the Groud Up, AK Peters, <http://www.raytracergroundup.com>

Matt Pharr, Greg Humphreys, Physically-based Rendering, Morgan Kaufmann, <http://www.pbrt.org>

Matthias Teschner, Advanced Computer Graphics, Course material, <http://cg.informatik.uni-freiburg.de/teaching.htm>